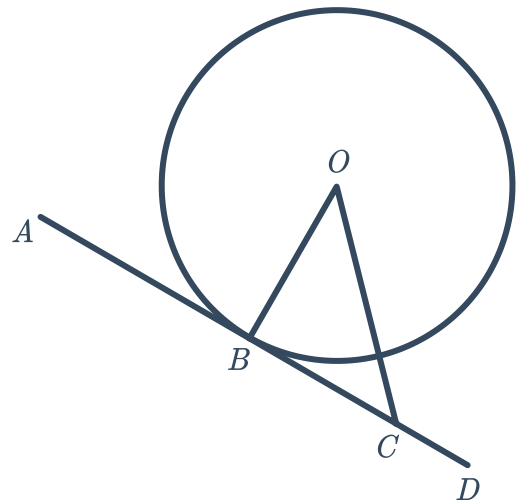


## Tangents to a circle



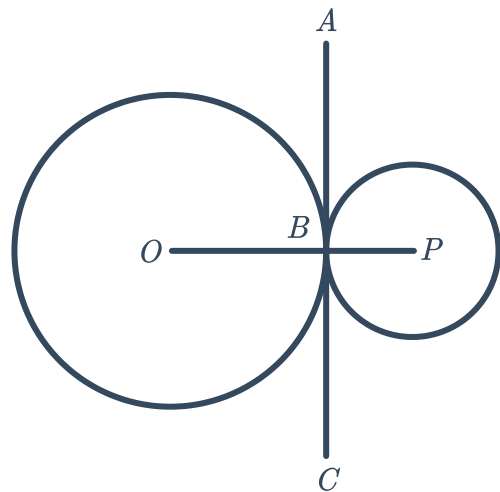
1 In the diagram,  $C$  is an arbitrary point on the line  $AD$ , and  $B$  is the point at which the tangent meets the circle.

- a What can be said about the lengths of  $OB$  and  $OC$ ?
- b What point on  $AD$  is closest to the centre of the circle?
- c In general, what can be said about the angle of a line joining some point to some other line by the shortest route?
- d Hence, what can be said about angle  $\angle OBA$ ?



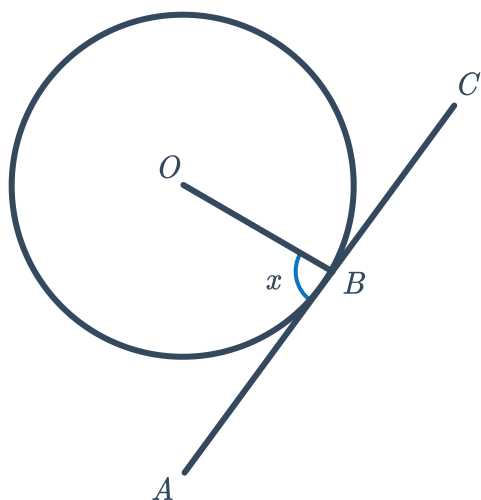
2 Consider the following figure:

- a Show that  $\angle OBP$  is a straight angle.
- b Hence what can we say about points  $O$ ,  $B$  and  $P$ ?

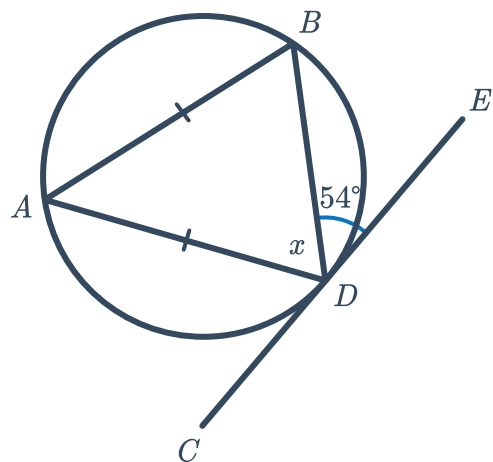


3 Find the value of the pronumerals in the following diagrams. Give reasons for your answer.

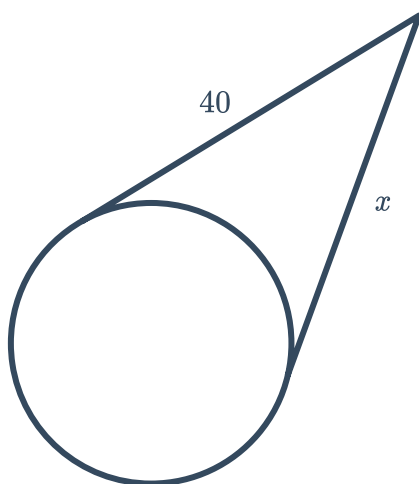
a



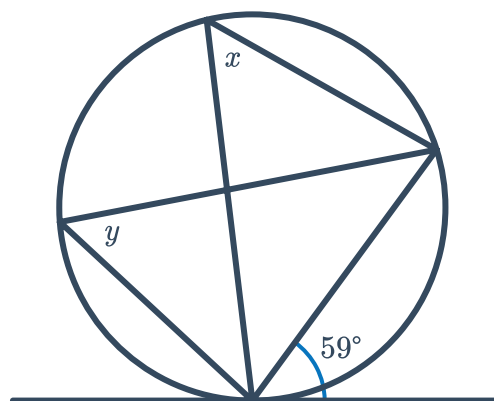
b



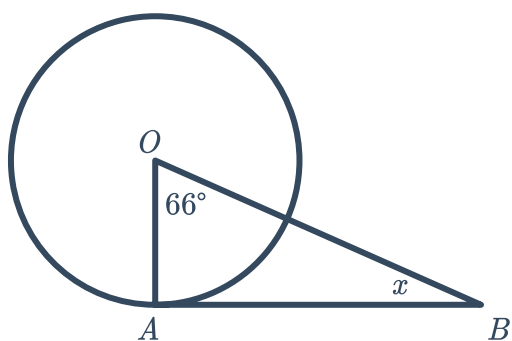
c



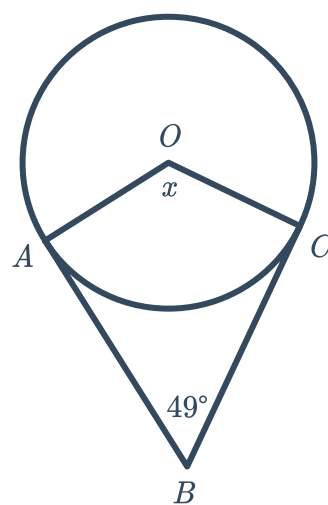
d



e



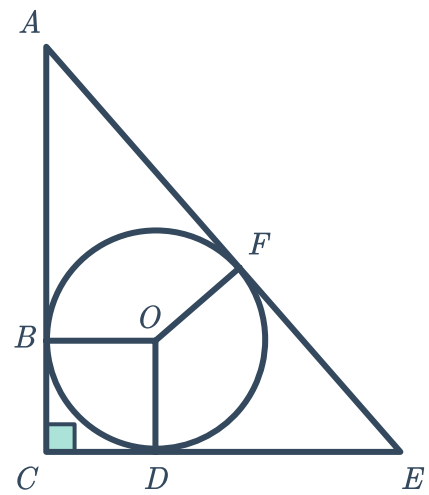
f



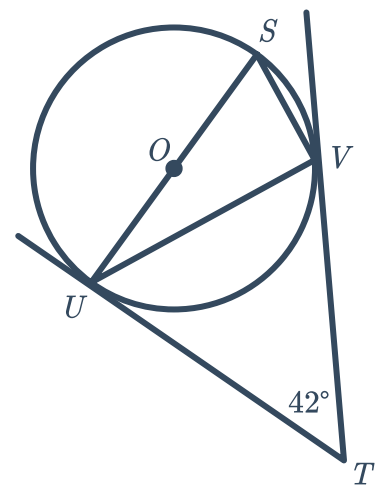
- 4 The circle in the following diagram has a radius of 8 and  $AB$  has a length of 9 units.



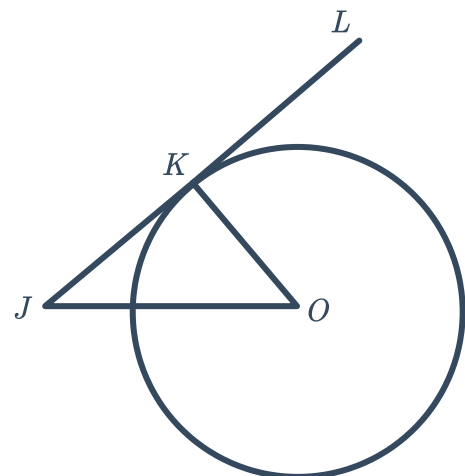
Calculate the length of  $FE$ .



- 5 In the diagram,  $TU$  and  $TV$  are tangents to the circle with centre  $O$ . Find the size of  $\angle SUV$ .



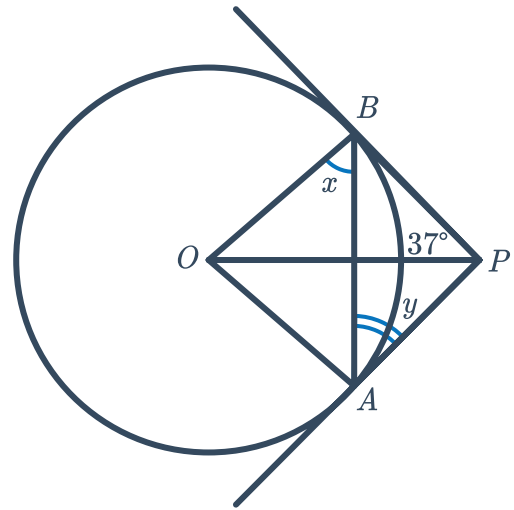
- 6 In the following diagram,  $JL$  is a tangent to the circle with centre  $O$ .  $JO = 29$  and  $JK = 20$ . Find the length of the radius  $OK$ .



- 7 In the following diagram,  $PA$  and  $PB$  are two tangents issued from point  $P$  to the circle of center  $O$ , such that  $\angle PBO = 37^\circ$ .

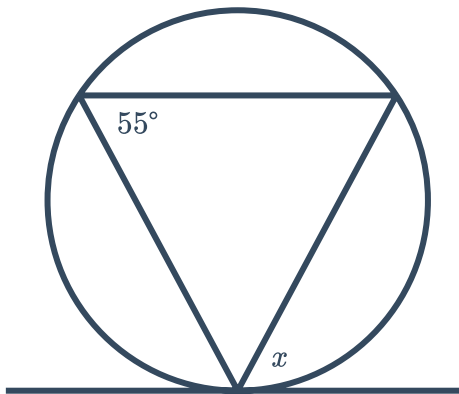


- a Find the value of  $y$ .  
b Find the value of  $x$ .

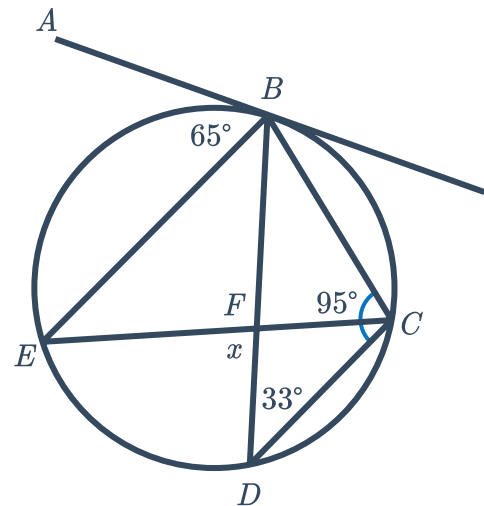


- 8 Find the value of the pronumerals in the following diagrams. Give reasons for your answer.

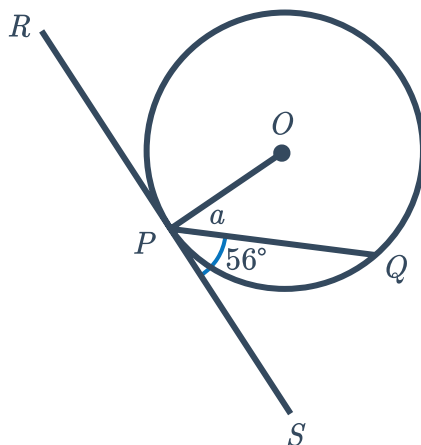
a



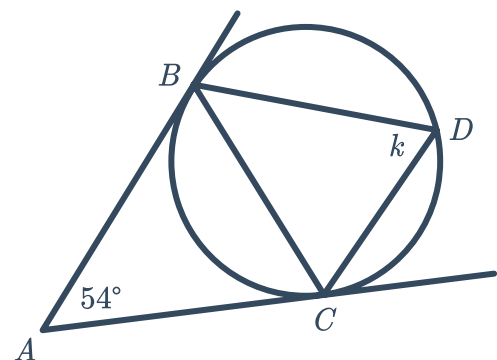
b



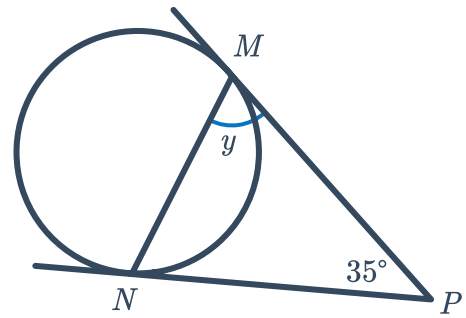
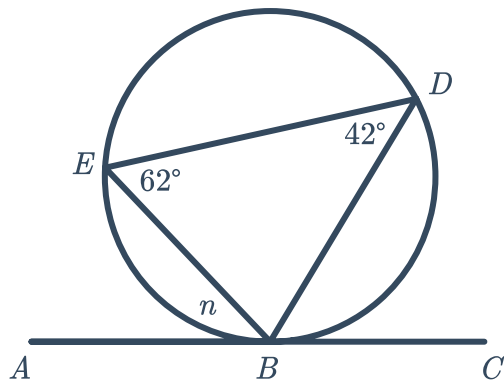
c



d

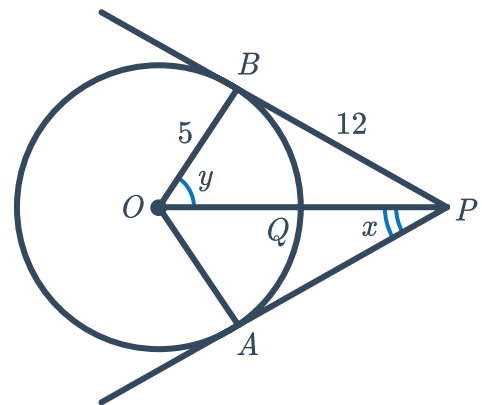


e

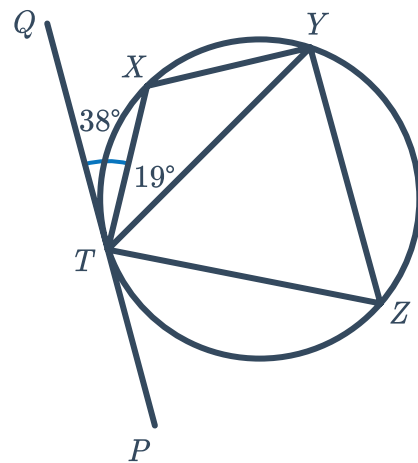


- 9 In the following diagram,  $PA$  and  $PB$  are two tangents issued from point  $P$  to the circle of center  $O$ , such that  $PB = 5$ .

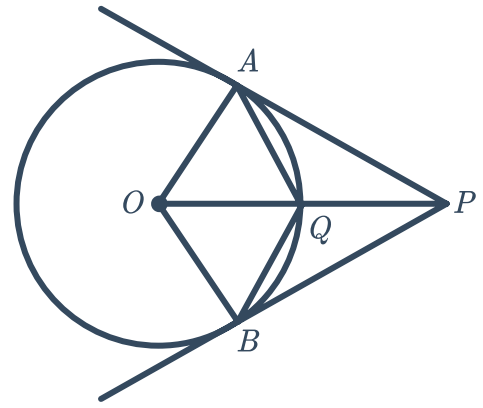
- a Find the value of  $x$ .  
b Find the value of  $y$ .



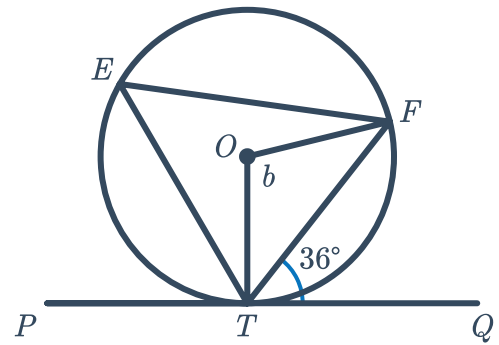
- 10 In the diagram,  $PT$  is a tangent to the circle. Find the size of  $\angle YZT$ .



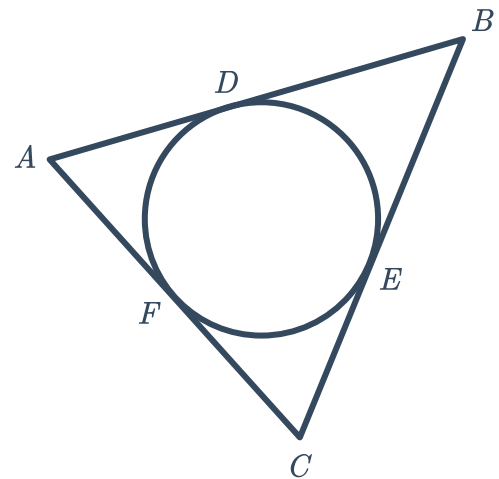
- 11 In the figure,  $O$  is the center of the circle,  $PA$  and  $PB$  are tangents. Find the size of  $\angle APB$  if  $\angle OAQ = 56^\circ$ .



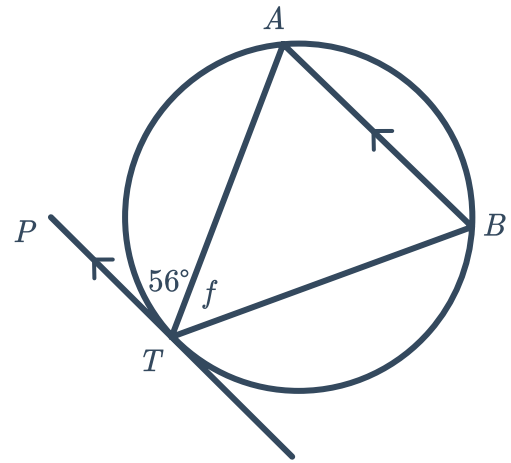
- 12 In the diagram,  $PT$  is a tangent to the circle with centre  $O$ . Determine the value of  $b$ .



- 13 In the following diagram,  $AB$ ,  $BC$  and  $AC$  are tangents to the circle.  $BD = 7$  cm,  $AF = 6$  cm, and  $BC = 11$  cm. Find  $p$ , the perimeter of  $\triangle ABC$ .

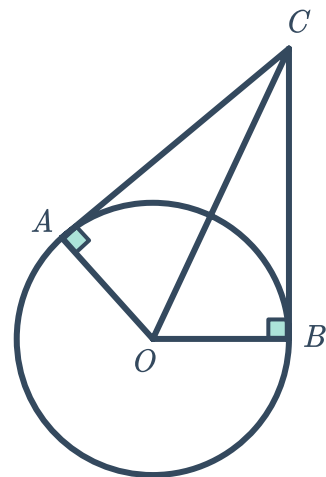


- 14 In the diagram,  $PT$  is a tangent to the circle.  
Determine the value of  $f$ .

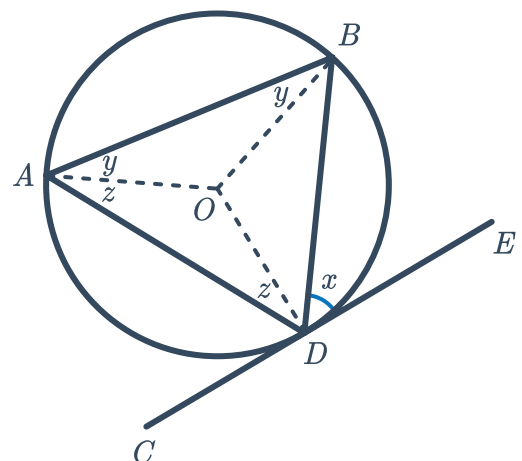


## Proofs

- 15 Prove that  $\triangle OAC$  and  $\triangle OBC$  are congruent. Then, show that  $AC = BC$ .



- 16 For the following diagram, prove that  $y + z = x$ .



- 17 For the following diagram, prove that  $AB \parallel FG$ .

